

THE WEST AFRICAN EXAMINATIONS COUNCIL

BASIC EDUCATION CERTIFICATE EXAMINATION -- JUNE 2008

INTEGRATED SCIENCE -- PAPER 1

Duration: 1 Hour | Level: JHS 3 / BECE | Code: SCI 01/02 - P1

INSTRUCTIONS TO CANDIDATES

Answer all 40 questions on your objective answer sheet. Each question is followed by four options lettered A to D. Choose the correct option and mark it with 2B pencil.

PART I: OBJECTIVE QUESTIONS

1. Which of the following cellular structures is present in plant cells but absent in animal cells?

- A. Cell membrane
- B. Cytoplasm
- C. Cellulose cell wall
- D. Mitochondria

2. The process by which green plants manufacture carbohydrates using sunlight is called...

- A. Transpiration
- B. Respiration
- C. Photosynthesis
- D. Fermentation

3. Which of the following is an example of a chemical change?

- A. Melting of ice
- B. Rusting of iron
- C. Dissolving sugar in water
- D. Evaporation of alcohol

4. The SI unit for measuring electrical resistance is...

- A. Ampere
- B. Volt
- C. Ohm
- D. Watt

5. Which of the following gas components makes up approximately 78% of atmospheric air?

- A. Oxygen
- B. Carbon dioxide
- C. Nitrogen
- D. Argon

6. A substance with a pH value of 3.0 is classified as...

- A. Strongly acidic
- B. Weakly acidic
- C. Neutral
- D. Basic

7. Which of the following organisms causes malaria in humans?

- A. Female Anopheles mosquito
- B. Plasmodium parasite
- C. Tsetse fly
- D. Housefly

8. The bending of a ray of light as it passes from air into water is known as...

- A. Reflection
- B. Refraction
- C. Dispersion
- D. Diffraction

9. Which part of the human alimentary canal is primarily responsible for the absorption of digested nutrients?

- A. Stomach
- B. Duodenum
- C. Ileum (Small intestine)
- D. Colon

10. An example of a first class lever is...

- A. Wheelbarrow
- B. Pair of scissors
- C. Nutcracker
- D. Fishing rod

11. The process of removing metabolic waste products from the body of a living organism is called...

- A. Egestion
- B. Excretion
- C. Secretion
- D. Defecation

12. Which of the following soil types has the highest water retention capacity?

- A. Sandy soil
- B. Loamy soil
- C. Clay soil
- D. Gravel

13. What is the chemical formula for ordinary table salt?

- A. NaOH
- B. NaCl
- C. CaCO₃
- D. HCl

14. The transfer of heat through solids from a region of higher temperature to lower temperature without movement of the material is...

- A. Conduction
- B. Convection
- C. Radiation
- D. Evaporation

15. Which vitamin is synthesized in the human skin upon exposure to morning sunlight?

- A. Vitamin A
- B. Vitamin B
- C. Vitamin C
- D. Vitamin D

16. The male reproductive organ of a flowering plant is known as the...

- A. Carpel
- B. Pistil
- C. Stamen
- D. Petal

17. Which of the following agricultural practices directly prevents soil erosion on steep hillsides?

- A. Terracing
- B. Slash and burn
- C. Monoculture
- D. Overgrazing

18. The density of a substance is defined as...

- A. Mass x Volume
- B. Mass / Volume
- C. Weight / Volume
- D. Volume / Mass

19. Which blood cell type is responsible for defending the human body against infectious pathogens?

- A. Red blood cells
- B. White blood cells
- C. Platelets
- D. Plasma

20. The chemical symbol for Potassium is...

- A. P
- B. Pt
- C. K
- D. Po

21. An echo is produced due to the...

- A. Refraction of sound
- B. Reflection of sound
- C. Absorption of sound
- D. Diffraction of sound

22. Which of the following disease conditions is caused by a deficiency of iodine in the human diet?

- A. Rickets
- B. Scurvy
- C. Goitre
- D. Beriberi

23. The process by which water vapour cools and turns into liquid droplets is called...

- A. Sublimation
- B. Evaporation
- C. Condensation
- D. Freezing

24. Which gas is evolved when dilute hydrochloric acid reacts with calcium carbonate?

- A. Oxygen
- B. Hydrogen
- C. Carbon dioxide
- D. Chlorine

25. What type of energy transformation occurs in a microphone?

- A. Electrical to sound
- B. Sound to electrical
- C. Mechanical to thermal
- D. Chemical to light

26. A safety device in an electrical circuit that melts to break the circuit when excessive current flows is a...

- A. Switch
- B. Fuse
- C. Resistor
- D. Transformer

27. Which planet is closest to the Sun in our Solar System?

- A. Venus
- B. Mercury
- C. Earth
- D. Mars

28. The loss of water in the form of water vapour from the aerial parts of a plant is termed...

- A. Guttation
- B. Transpiration
- C. Osmosis
- D. Diffusion

29. Which of the following instruments is used to measure atmospheric pressure?

- A. Thermometer
- B. Barometer
- C. Hydrometer
- D. Anemometer

30. What is the chemical name of rust?

- A. Hydrated iron(III) oxide
- B. Iron(II) chloride
- C. Iron(II) sulphate
- D. Iron sulphide

31. Which organ in the human body filters urea from the bloodstream to form urine?
 - A. Liver
 - B. Kidney
 - C. Lungs
 - D. Heart
32. The process of heating milk to kill disease-causing microorganisms is called...
 - A. Sterilization
 - B. Pasteurization
 - C. Fermentation
 - D. Refrigeration
33. In magnetism, like poles...
 - A. Attract
 - B. Repel
 - C. Neutralize
 - D. Do not interact
34. Which of the following is a non-renewable source of energy?
 - A. Solar energy
 - B. Crude oil
 - C. Wind energy
 - D. Biomass
35. The component of a flower that attracts insect pollinators is the...
 - A. Sepal
 - B. Petal
 - C. Receptacle
 - D. Filament
36. The boiling point of pure water at standard atmospheric pressure is...
 - A. 0 deg C
 - B. 50 deg C
 - C. 100 deg C
 - D. 212 deg C
37. Which gas is essential for human aerobic cellular respiration?
 - A. Carbon monoxide
 - B. Nitrogen
 - C. Oxygen
 - D. Helium
38. A mixture of iron filings and sulphur powder can be separated physically by using a...
 - A. Filter paper
 - B. Magnet
 - C. Separating funnel
 - D. Condenser
39. The female gamete in humans is produced in the...
 - A. Uterus
 - B. Ovary
 - C. Fallopian tube
 - D. Vagina
40. Which of the following is a biodegradable waste material?
 - A. Polythene bag
 - B. Glass bottle
 - C. Banana peel
 - D. Aluminium can

OFFICIAL WAEC ANSWER KEY & SCORING GUIDE

For candidate self-assessment and teacher evaluation:

1. [C] 2. [C] 3. [B] 4. [C] 5. [C]
6. [A] 7. [B] 8. [B] 9. [C] 10. [B]
11. [B] 12. [C] 13. [B] 14. [A] 15. [D]

16. [C] 17. [A] 18. [B] 19. [B] 20. [C]
21. [B] 22. [C] 23. [C] 24. [C] 25. [B]
26. [B] 27. [B] 28. [B] 29. [B] 30. [A]
31. [B] 32. [B] 33. [B] 34. [B] 35. [B]
36. [C] 37. [C] 38. [B] 39. [B] 40. [C]